

Wireless Hacking

1. Wireless Network Basics

A **wireless network (Wi-Fi)** allows devices to communicate using **radio frequency signals** instead of cables.

Key components:

- **Access Point (AP)** – Device that provides wireless connectivity.
- **Client/Station** – Devices like laptops or phones connecting to the AP.
- **SSID (Service Set Identifier)** – The Wi-Fi network name.
- **BSSID** – MAC address of the access point.
- **Channel** – Frequency used for communication.

Protocols commonly used:

- **WEP**
 - **WPA**
 - **WPA2**
 - **WPA3**
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2. Wireless Security Weaknesses

Wireless networks are vulnerable because **signals travel through the air**, meaning attackers only need to be **within range**.

Common weaknesses include:

1. Weak Encryption

Older protocols like **WEP** use weak encryption and can be cracked quickly.

Problems:

- Small initialization vectors
 - Reused keys
 - Weak RC4 encryption
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2. Weak Passwords

Networks using **WPA2-PSK** can be attacked if the password is weak.

Attack concept:

1. Capture handshake
 2. Perform dictionary attack
 3. Crack the password offline
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3. MAC Address Spoofing

Every device has a **MAC address**.

Attackers can:

- Change their MAC address
- Bypass MAC filtering
- Impersonate legitimate devices

Example concept:

Original MAC: 00:11:22:33:44:55

Spoofed MAC: 90:AB:CD:EF:12:34

3. Wireless Reconnaissance

Before exploitation, attackers perform **wireless reconnaissance**.

Goals:

- Identify networks
- Detect encryption type
- Find clients connected to APs

Methods:

- **Passive scanning** (listening to packets)
- **Active scanning** (sending probe requests)

Common information collected:

- SSID
 - BSSID
 - Channel
 - Encryption type
 - Signal strength
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4. Packet Sniffing

```
sudo airodump-ng -i wlan1mon  
sudo airodump-ng --bssid 62:FA:C4:DB:E0:24 -c 10 -w /tmp/Zone4  
wlan1mon
```

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Wireless networks broadcast packets that can be captured.

Concept:

- Capture data frames
- Analyze traffic
- Identify authentication handshakes

Attackers often target:

- **WPA2 Four-Way Handshake**

This handshake is used to verify authentication between a client and AP.

5. Deauthentication Attacks

```
sudo aireplay-ng --deauth 1 -a 42:6E:82:D6:BB:80 -c  
3C:55:76:69:F3:C3 wlan1mon
```

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-a bssid : set Access Point MAC address

-c dmac : set Destination MAC address

--deauth 0 for unlimited

A **deauthentication attack** forces a client to disconnect from the network.

Concept:

1. Send fake **deauth frames*
2. Client disconnects
3. Client reconnects
4. Attacker captures authentication handshake

Weakness:

Early **IEEE 802.11** standards did not protect management frames.

Cracking

```
# CPU  
aircrack-ng /tmp/Zone4-01.cap -w /usr/share/wordlists/rockyou.txt  
  
# hashcat GPU  
hcxpcapngtool -o Zone4.22000 /tmp/Zone4-01.cap  
hashcat -m 22000 Zone4.22000 /usr/share/wordlists/rockyou.txt
```

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6. Evil Twin Attack

An **Evil Twin** is a fake access point.

Concept:

1. Attacker creates a fake AP with same SSID
2. Victim connects to it
3. Attacker intercepts traffic

Used for:

- Credential harvesting
 - Phishing
 - Man-in-the-Middle attacks
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7. Man-in-the-Middle (MITM)

In wireless attacks, attackers place themselves between the **victim and router**.

They can:

- Intercept traffic
- Modify packets
- Redirect users to fake pages

Example techniques:

- ARP poisoning
 - Rogue access points
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8. Rogue Access Points

A **rogue AP** is an unauthorized wireless access point connected to a network.

Risks:

- Internal network access
 - Data interception
 - Credential theft
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9. Wireless Denial of Service (DoS)

Attackers can disrupt Wi-Fi networks.

Methods:

- Deauthentication flooding
- RF jamming
- Association request flooding

Goal:

- Disconnect users from the network
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10. Post-Exploitation in Wireless Attacks

After gaining access, attackers may attempt:

- Network scanning
 - Credential harvesting
 - Traffic interception
 - Lateral movement in LAN
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11. Wireless Security Best Practices

To protect networks:

- Use **WPA3**
- Strong password
- Disable WPS
- Use network segmentation
- Enable **802.11w Protected Management Frames**
- Monitor rogue APs